

Affan School Management System

Software Requirements Specification Document



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Affan School Management System

Revision History

Date	Description	Author	Comments
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Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Signature	Printed Name	Title	Date
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Chapter No: 01

Introduction

Affan School Management Systems Plays an essential role in the current educational system. School authorities all over the world are engaged in a lot of day-to-day administrative and academic activities to manage and provide a better academic experience to students effectively. School ERP is a web based School Management System that enables school to use and operate many of integrated interrelated modules and manage the administration of school efficiently. **Affan school management system helps the parents monitor their children from anywhere. They can check their children's academic performance from a remote location.**

1.2 Scope

E-School Management System is comprehensive web-based School Management Software. It will design for better interaction between students, teachers, parents & management.

1.3 Definitions, Acronyms, and Abbreviations.

This document is formed using IEEE SRS format, headings are in bold capital letters and wherever necessary diagram is provided.

- ASMS - Affan school management system
- OSMS - Online school Management System
- PPEE - Php Platform Enterprise Edition
- SQL - sql Server Page
- SRS - Software Requirements Specification
- OS - Operating System

1.4 References

https://www.academia.edu/24648944/SOFTWARE_REQUIREMENTS_SPECIFICATION_SR_FOR_STUDENT_INFORMATION_MANAGEMENT_SYSTEM

--Software Engineering a Practitioner's Approach by Roger S. Pressman

--Class Power Point presentation slides.

--For Database sql

1.5 Overview

Affan School Management System can provide following benefits in general to your school, college or institute:

- Single system to manage all School related information from anywhere in the School.
- Easy to use school ERP
- Reliable and secure system Complete
- Automation of operations More Time to focus on Strategic Tasks
- Better informed decision making for management
- Parents have access to all academic information about their wards through the internet

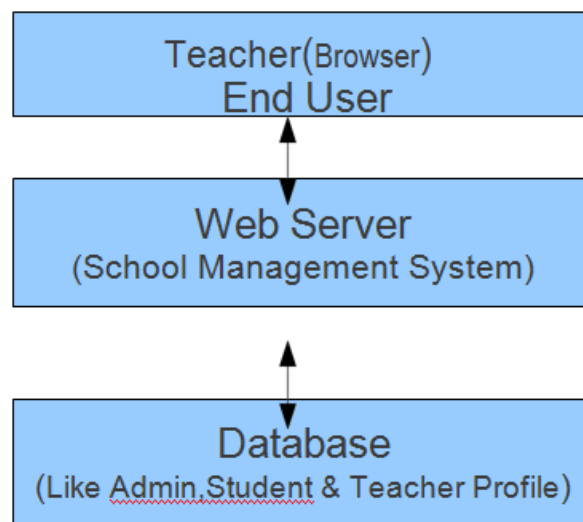


Figure 1.5 Overview

Genral Description

The student management system allows authorized members to access the records of academically registered students. It can be used in various educational institutes across the globe and simplifies working of institutes

2.1 Product Perspective

The proposed system shall be developed using client/server architecture and be compatible with Microsoft Windows Operating System. The front end of the system will be developed using Dreamweaver and backend will be developed using MS SQL Server 2000.

2.1.1 Operations

- i. The ONSMS will have following user-friendly and menu driven interfaces
- ii. Login: to allow the entry of only authorized users through valid login Id and password.
- iii. School Details: to maintain school details.
- iv. Program Details: to maintain program details.
- v. Scheme Details : to maintain scheme details of a school
- vi. Paper Details: to maintain paper details of a scheme for a particular program
- vii. Faculty Details : to maintain the faculty details.

2.1.2 Site Adaptation Requirements

2.2 Product Functions

The OSMS will allow access only to authorized users with specific roles (System administrator, Faculty and Teacher). Depending upon the user's role, he/she will be able to access only specific modules of the system.

A summary of major functions that the URS will perform

- A login facility for enabling only authorized access to the system.

- System administrator will be able to add, modify or delete program, school, scheme, paper and login information.
- Students will be able to add/modify his/her details and register for papers to be studied in the current semester.
- System administrator/Faculty will be able to generate reports.

2.3 User Characteristics

Qualification: At least matriculation and comfortable with English.

- Experience: Should be well versed/informed about the registration process of the university.
- Technical Experience: Elementary knowledge of computers

2.4 General Constraints

There will only be one administrator.

- The delete operation is available only to the administrator. To reduce the complexity of the system, there is no check on delete operation. Hence, administrator should be very careful before deletion of any record and he/she will be responsible for data consistency.

2.5 Assumptions and Dependencies

- The login Id and password must be created by system administrator and communicated to the concerned user confidentially to avoid unauthorized access to the system.
- It is assumed that a student registering for the subsequent semester has been promoted to that semester by the university as per rules and has paid desired university fee.
- Registration process will be open only for specific duration

Chapter No: 03

Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

In the very first step the manager will authenticate his/herself. Then main page will open where all categories of available staff and students are in Recycler View of Home. User After login , add in to staff and checking out of a confirmation pop up message will appear.

3.1.2 Interfaces

- First of all screen shows Sign In(For account holder) and Sign Up(For new User) buttons then move to food page.
- As described above food page will consists of information and photographs about all the available categories of foods and after clicking on done button a confirmation pop up message will appear.
- When admin/user give ID and password and click the login button the main page will open and login form will disappear.

3.1.3 Hardware Interfaces

- a) Screen resolution of at least 640 x 480 or above.
- b) Computer systems will be in the networked environment as it is a multi-user system.

3.1.4 Software Interfaces

- a) MS-Windows Operating System
- b) Operating System: Windows 7(64bit)
- c) Adobe Dreamweaver CS
- d) Microsoft Dreamweaver for designing front-end
- e) MS SQL Server 2000 for backend
- f) PLATFORM : PHP LANGUAGE
- g) INTEGRATED DEVELOPMENT ENVIRONMENT(IDE):ECLIPSE

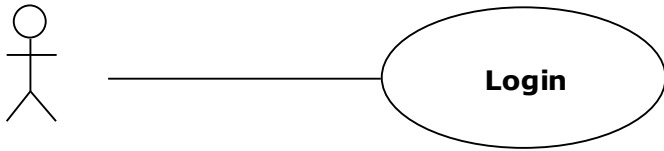
3.1.5 Communications Interfaces

- None

3.2 Functional Requirements

- Our system provides these facilities.
- Store add staff and student detail in database.
- Our system provides a feature to update database in future as required.
- The system saves the record of the staff, student, exam and results.
- Admin must provide ID and Password before interacting with the system for authorization purpose.
- System must provide security from unauthorized access.

3.3 Use Cases 1

3.3.1 Use Cases UC-01	
Use Case Name	Login
Actors	Administrator/User
Use Case Diagram	
Description	This use case describes the process of login verification for the users of the system
Pre-Condition	Database should be connected.
Primary Scenario	Step 1. The user will provide username and password Step 2. The user will click login. Step 3. The system will validate the login information. Step 4. The system will show the success message.

	Step 5. The system will open the main application.
Secondary Scenario	1. The administrator can click exit to close the application.
Exceptions	Step 2. Username/Password not supplied. Step 3. Username/Password is not valid. Step 5. Unable to connect to database, the application will be closed.
Post-Condition	The system shall apply the proper access rights defined for the current user.

3.3.2 Use Cases 2

Use Case ID UC-02	
Use Case Name	Admin
Actors	Administrator
Use Case Diagram	<pre> graph LR Actor(()) --- UC((Manage Users)) </pre> The diagram shows a stick figure actor connected by a line to an oval use case labeled 'Manage Users'.
Description	Administrator can Save, Update, View or Delete user information.
Pre-Condition	Administrator must be logged in.
Primary Scenario	Step 1. Administrator will click on view User information. Step 2: System will display User information form Step 3. The administrator would browse for the existing information of user. Step 4. The system will display its preview. Step 5. The administrator will save, update, delete user information

	Step 6. The system will take action according to select operation. Step 7. The administrator will click save. Step 3. The system will save the user information and display success message. Step 4. The Administrator will select click update. Step 5. The system will display all the information of the user. Step 6. The administrator will change the information and clicks save to update the user information. Step 7. The system will save the user information and display success message. Step 8. The administrator will click delete button. Step 9. The system will delete the selected user information and displays success message.
Secondary Scenario	1. The administrator can click exit to leave the process. 2. The administrator can come back to browse another user information.
Exceptions	Step 10. (a) Information not supplied. (b) Information not valid. Step 11. Invalid data entry in fields do not perform selected operation successfully.

3.4 Classes / Objects

- This section contains major categories of the “Taste It”.

a) <Admin>

b) ID, Password

c) for login for interaction with system

d) <User>

e) Name, Cell number Mail

f) for storing information of user

g) <Exam>

h) Category

i) for storing semester session categories

3.5 Non-Functional Requirements.

3.5.1 Performance: Working will be fast.

3.5.2 Reliability: System will be easy to operate.

3.5.3 Availability: The system is easy to use because GUI is provided.

3.5.4 Security: System will restrict unauthorized access.

3.5.5 Maintainability: System will be easy to update as needed.

3.5.6 Portability: System will be easy to use on any Desktop Computer.

3.6 Inverse Requirements

- class only be selected from our given categories.

3.7 Logical Database Requirements

- Database must be valid for the system.
- The database must possess all the record of our system.
- Teacher is not allowed to interfere with the database.
- There will be different tables in database Admin authentication, user records and Student category record.

3.8 Design Constraints

- All fields must be filled correctly
- User's data will automatically be stored in firebase database on user sign up.
- After completing an operation display a message of record save or error.
- All necessary fields must be filled

Standards Compliance

1. Perform a system safety or security assessment
2. Determine a target system failure rate
3. Use the system target failure rate to determine the appropriate level of development rigor
4. Use a formal requirements capture process
5. Create software that adheres to an appropriate coding standard
6. Trace all code back to their source requirements
7. Develop all software and system test cases based on requirements
8. Trace test cases to requirements
9. Use coverage analysis to assess test completeness against both requirements and code
10. For certification, collect and collate the process artifacts required to demonstrate that an appropriate level of rigor has been maintained.

Analysis Models

4.1 Sequence Diagrams

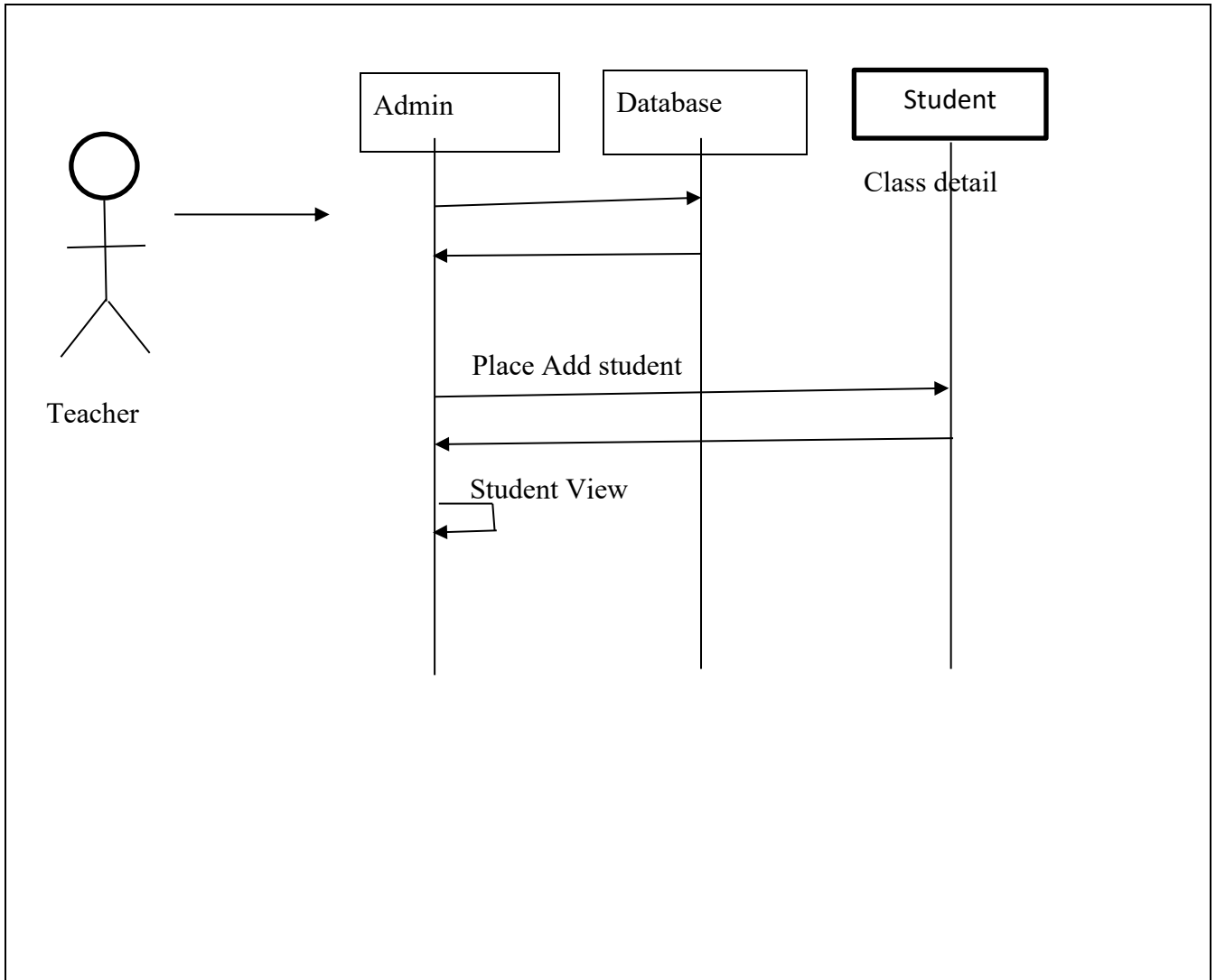


Figure 4.1.1 Sequence Diagrams

4.2 Data Flow Diagrams

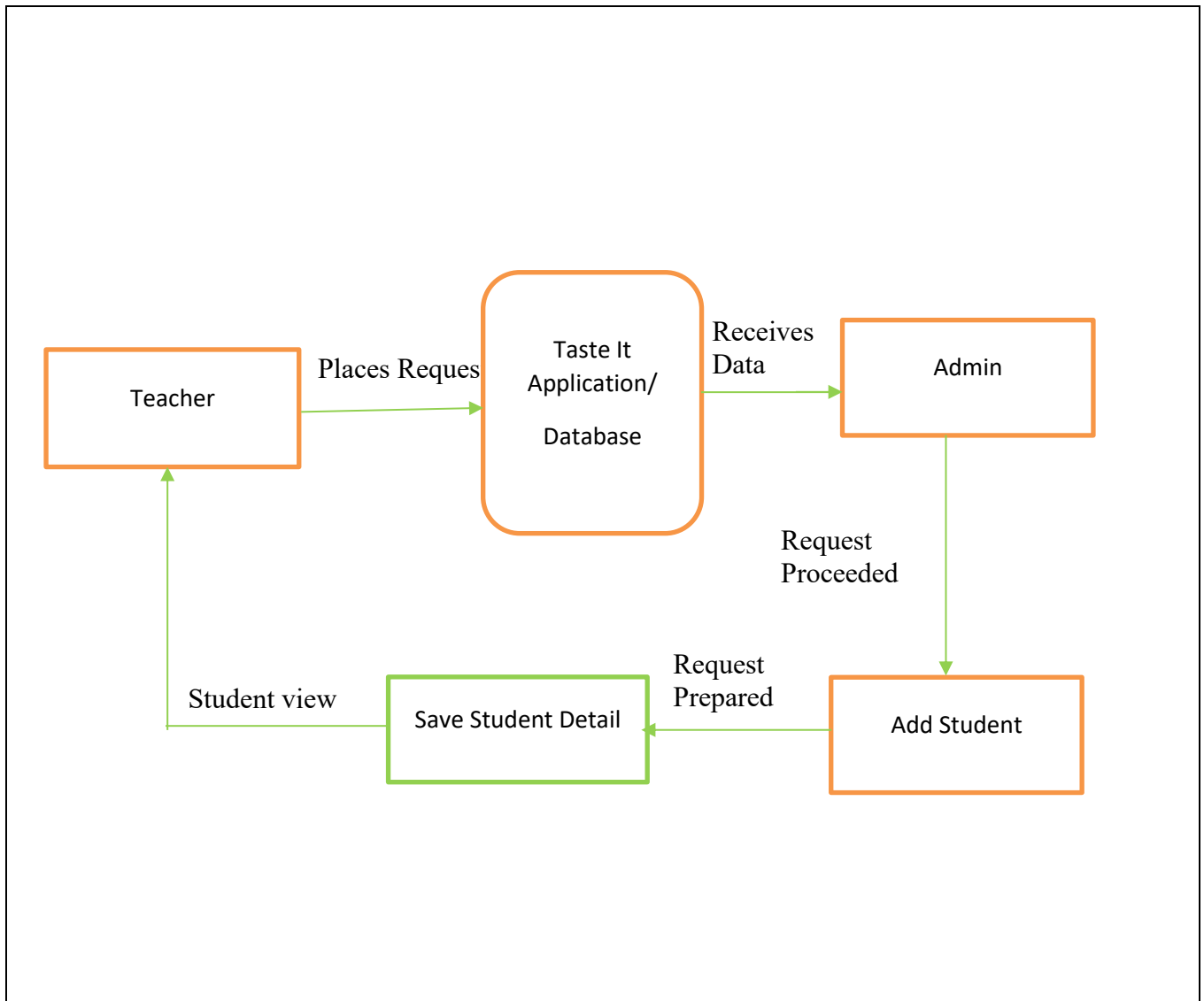


Figure 4.2.1 Data Flow Diagrams

4.3 State-Transition Diagrams

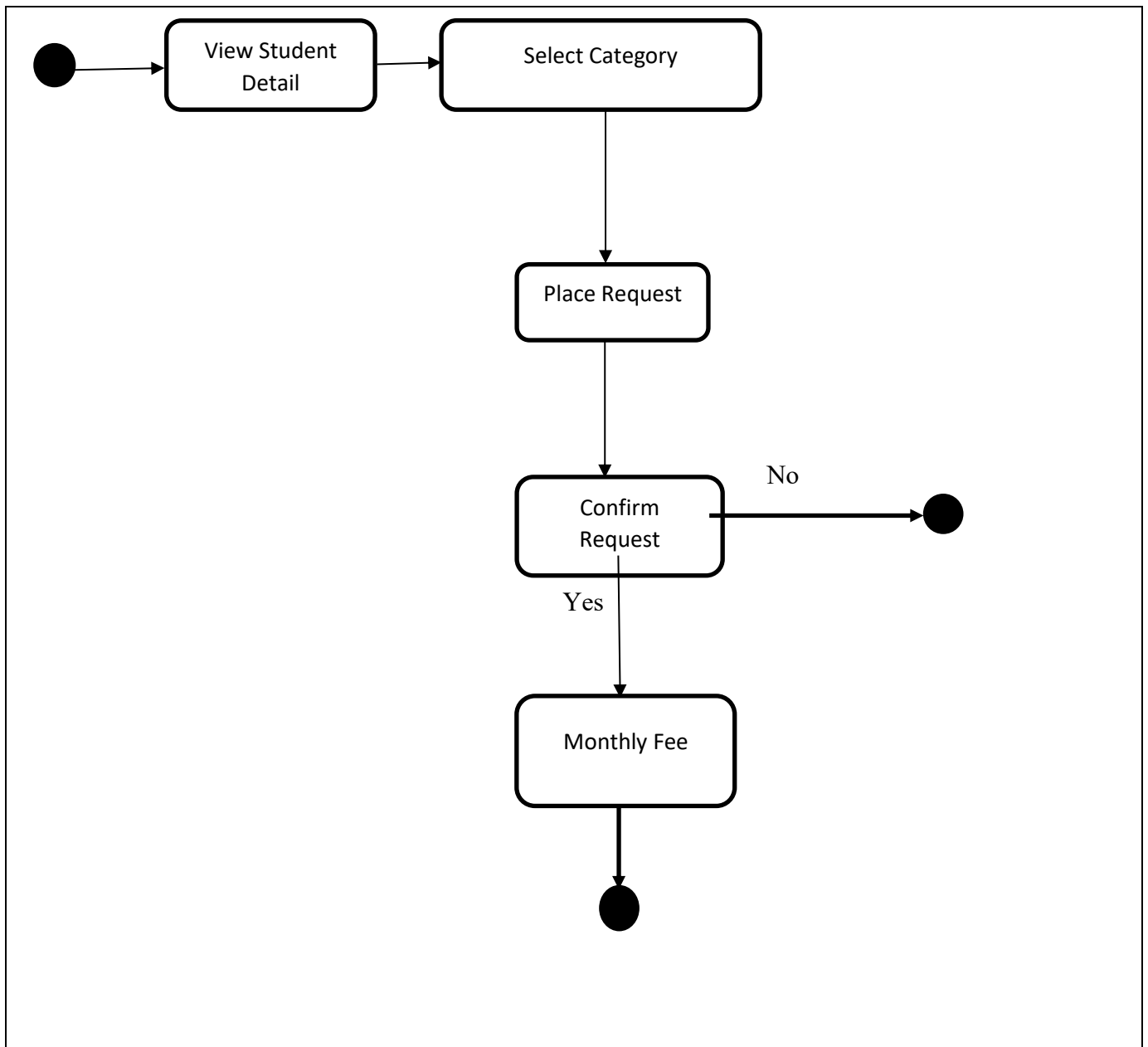


Figure 4.3.1 State-Transition Diagrams

Appendices

5.1 Appendices

A: Admin, Abbreviation, Acronym, Assumptions;

B: Books, Business rules;

C: Class, Client, Conventions;

D: Data requirement, Dependencies;

G: GUI;

K: Key;

L: Library, Librarian;

M: Member;

N: Non-functional Requirement;

O: Operating environment;

P: Performance, Perspective, Purpose;

R: Requirement, Requirement attributes;

S: Safety, Scope, Security, System features;

U: User, User class and characteristics, User requirement;

Glossary

- The following are the list of conventions and acronyms used in this document and the project as well:
- Admin: A login id representing a user with user administration privileges to the software
- student: A general login id assigned to most users
- Client: Intended users for the software
- SQL: Structured Query Language; used to retrieve information from a database
- SQL Server: A server used to store data in an organized format
- Layer: Represents a section of the project
- User Interface Layer: The section of the assignment referring to what the user interacts with directly
- Application Logic Layer: The section of the assignment referring to the Web Server. This is where all computations are completed
- Data Storage Layer: The section of the assignment referring to where all data is recorded
- Use Case: A broad level diagram of the project showing a basic overview
- Class diagram: It is a type of static structure diagram that describes the structure of a system by showing the system's cases, their attributes, and the relationships between the classes
- Interface: Something used to communicate across different mediums
- Unique Key: Used to differentiate entries in a database